# Shiv Nadar University, Chennai

# Department of Information Technology

# 25IT1731 Computational Thinking and Problem Solving with Python

## Ex 2 - Control statements

## Part A

Write Python programs for all the following exercises using conditional statements and loops.

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| 1 | Write a program that reads the three sides of a triangle. First check whether the sides can actually form a valid triangle (the sum of any two sides must be greater than the third side). If valid, classify the triangle as equilateral, isosceles, or scalene; if not valid, display an appropriate message. |
| 2 | A firm awards a festival bonus to its staff. Male employees receive a 6% bonus on their salary, while female employees receive a 12% bonus. Additionally, if an employee's salary is below Rs. 12,000/-, they receive an extra 3% bonus. Write a program that computes the bonus amount and prints the final salary after the bonus is added. |
| 3 | Ask the user to enter marks scored in four subjects. Compute the total and percentage, then display a grade based on the following criteria:<br>percentage ≥ 90 -- Grade A<br>80 ≤ percentage < 90 -- Grade B<br>70 ≤ percentage < 80 -- Grade C<br>60 ≤ percentage < 70 -- Grade D<br>percentage < 60 -- Fail |
| 4 | Write a program that takes a year as input and determines whether it is a leap year, applying the rule that a year divisible by 4 is a leap year unless it is a century year, in which case it must also be divisible by 400. |
| 5 | Write a program using the match-case statement that reads an integer from 1 to 12 and prints the corresponding month name. If the input is outside this range, print “Invalid month”. |
| 6 | Write a program that reads a sentence from the user and counts how many vowel it contains. |
| 7 | Write a program to compute the sum of each of the following series up to n terms:<br>a) $2 + 4 + 6 + \cdots +$ n terms<br>b) $1 + x^2/2 + x^3/3 + \cdots + x^n/n$<br>c) $x - x^2/2 + x^3/3 - x^4/4 + \cdots$ up to n terms |
| 8 | Write a Python program to read a number and check whether it is a palindrome, using a while loop. (For example, 121 and 1331 are palindromes, while 123 is not.) |
| 9 | Write a program to print all the prime numbers between 1 and 500. |
| 10 | Write a Python program using a while loop that repeatedly asks the user to enter a number. For each number entered, the program should update running counts of: how many numbers were even, how many were odd, how many were positive, and how many were negative. The user should be able to type “exit” at any point instead of a number to stop entering values, after which the program should display all four counts along with the total number of values entered. |

## Part B

Write Python programs for the following data analysis problems using loops.

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| 11 | A digital payment application records the UPI transactions made by a user during a day. The user performs n transactions, and the transaction amount for each transaction is entered.<br>Write a Python program using a loop to calculate:<br>1. Total amount spent<br>2. Average transaction amount<br>3. Highest transaction amount<br>4. Lowest transaction amount<br>5. Number of transactions exceeding ₹5,000<br>The program should process each transaction amount using a loop. |
| 12 | A food delivery app collects customer ratings (from 1 to 5) for *n* orders delivered in a day.<br>Write a Python program using a loop to calculate:<br>1. Total number of ratings received,<br>2. Average rating, Number of 5-star ratings<br>3. Number of ratings below 3 (considered "poor service")<br>4. Percentage of orders rated 4 or above |
| 13 | Write a Python program to simulate a simple ATM machine using a while loop. The program should start by asking the user to set up an account with an initial balance and a 4-digit PIN. The user must enter the correct PIN before accessing the menu. The program should repeatedly allow the user to choose from:<br>1. Check Balance<br>2. Deposit Money<br>3. Withdraw Money (do not allow withdrawal if it exceeds the available balance; also limit a single withdrawal to a maximum of ₹50,000)<br>4. Change PIN<br>5. Exit<br>The program should keep running until the user selects Exit, and should display suitable error messages for invalid choices, incorrect PIN entries, or insufficient balance. |

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